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PROFILE

PhD candidate in applied mathematics working at the interface of inverse problems, computational imaging, and machine learning. My research designs fast, large-scale reconstruction methods for 3D photoacoustic tomography (PAT), carried out jointly between the Institut de Mathématiques de Toulouse (France) and A*STAR / IPAL (Singapore). Expected thesis defense: spring 2028.

CURRENT POSITION

2024 - 2028 **PhD Candidate in Applied Mathematics**

*IMT, Université de Toulouse, France & BII, A*STAR / IPAL, Singapore.*

RESEARCH OVERVIEW

Large-scale reconstruction methods for 3D photoacoustic tomography (PAT)

*PhD thesis — IMT Toulouse & A*STAR / IPAL Singapore, 2024–2028. Supervisors: C. Chaux, P. Escande, H. K. Lee.*

- Designing fast, accurate reconstruction methods — combining inverse problems, optimization, and deep learning — to deploy a novel 3D PAT scanner for biomedical imaging.
- Developed a combined Total Variation–Cauchy regularization for PAT image reconstruction (published at GRETSI 2025).
- Modeling the electrical impulse response of the 3D PAT system and implementing efficient forward/adjoint operators for large-scale data.
- In collaboration with the A*STAR team, exploring deep learning–based reconstruction for large-scale 3D PAT data.

Inverse problems in photoacoustic tomography

Research internship — IMT Toulouse, France, 2023. Supervisor: P. Escande.

- Implemented optimization algorithms to solve inverse problems in PAT.
- Built a blood-vessel image simulator for synthetic training data and trained deep neural networks for reconstruction (PyTorch).

Computer vision for road-safety analysis

Research internship — Cerema Occitanie, France, 2024. Supervisors: G. Saint-Pierre, P. Salmane.

- Analyzed road-user behavior from aerial / smart-camera data using object detection and trajectory extraction.
- Built a computer-vision pipeline for road-safety risk analysis (Python, deep learning); co-authored a paper at IEEE ICPRS-24.

EDUCATION

- 2024 - 2028** **PhD in Applied Mathematics** (in progress), Université de Toulouse, France — joint program with A*STAR, Singapore.
Inverse problems, computational imaging, and deep learning for 3D photoacoustic tomography.
- 2021 - 2024** **Engineering Degree (Master 2) in Applied Mathematics**, INSA Toulouse, France.
Department of Mathematical & Modeling Engineering (GMM).
Focus on statistics, optimization, and machine learning; coursework in inverse problems, signal/image processing, and high-dimensional data analysis.
- 2019 - 2021** **Preparatory Classes for Grandes Écoles (CPGE)**, Hue University of Education, Vietnam.
Two-year intensive program in mathematics and physics, preparing for the competitive entrance examinations of French engineering schools.
- 2016 - 2019** **Baccalaureate in Mathematics**, Quoc Hoc - Hué High School for the Gifted, Vietnam.
Specialized mathematics track; graduated with honors.

PROFESSIONAL EXPERIENCE

- May - June 2025** **Visiting Research Stay — Photoacoustic Image Reconstruction**
*BII, IPAL, A*STAR, Singapore.*
- Mar - Aug 2024** **Internship — Image Processing & Computer Vision**
Cerema Occitanie, France.
- June - Sept 2023** **Internship — Image Processing & Inverse Problems**
Institut de Mathématiques de Toulouse (IMT), France.

TEACHING EXPERIENCE

- 2025 - 2026** **Teaching Assistant — Optimisation (M1 Graduate Course)**
Université Toulouse III - Paul Sabatier (24h TD, 12h TP)
Led tutorials and lab sessions on continuous and convex optimization: gradient and descent methods, constrained optimization and KKT conditions, with numerical implementation in Python.
- 2024 - 2025** **Teaching Assistant — Fourier Analysis and Signal Theory (L3 Undergraduate Course)**
Université Toulouse III - Paul Sabatier (30h TD)
Led tutorials on Fourier series and transforms, sampling and convolution, and their application to signal filtering and spectral analysis.

PUBLICATIONS & PRESENTATIONS

Publications

- T.-T. Do**, C. Chaux, P. Escande, J. Gâteau, “Reconstruction d’images en tomographie photoacoustique avec régularisation combinée variation totale - Cauchy,” *Published at XXXe Colloque GRETSI 2025*.
- J. M. Rico Velázquez, P. H. Salmane, G. Saint-Pierre, L. Khoudour, **T. T. Do**, E. Martin, “Evaluation of road users trajectories from aerial-view data for risk analysis,” *Published at 14th International Conference on Pattern Recognition Systems (ICPRS), 2024*.

In Preparation

1. **T-T. Do**, et al., “Implementations of photoacoustic tomography models,” (*in preparation*).
2. **T-T. Do**, et al., “Electrical impulse response for 3D photoacoustic tomography models,” (*in preparation*).

Oral Presentations & Posters

- **Poster**, *Large-scale reconstruction methods for high-quality 3D photoacoustic imaging*, SinFra 2026, Singapore (Jun. 2026).
- **Poster**, *Journées modèles proxy et co-conception*, Institut Henri Poincaré, Paris, France (Sept. 2025).
- **Talk**, *XXXe Colloque GRETSI*, Strasbourg, France (Aug. 2025).

TECHNICAL SKILLS & LANGUAGES

<i>Programming</i>	Python, C/C++, CUDA, MATLAB, R.
<i>Scientific computing</i>	PyTorch, NumPy/SciPy, Scikit-Learn, Matplotlib.
<i>Tools</i>	Git, Docker, Linux, L ^A T _E X, Jupyter.
<i>Languages</i>	Vietnamese (native), English (professional), French (fluent).

OTHER INTERESTS

<i>Sports</i>	Football, racket sports, running.
<i>Music</i>	Guitar, pop music.
<i>Other</i>	Travelling, photography.